

Evolution of Digital Electronics

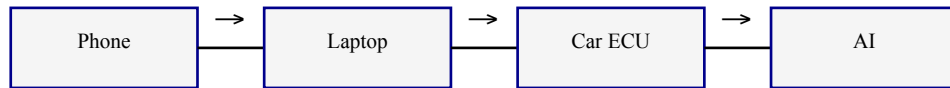
Reverse Engineering a Modern Chip

PowerPoint-style slide deck (PDF)

1. Question

What is inside a smartphone, laptop, car ECU, or AI server?

Answer: A digital chip.



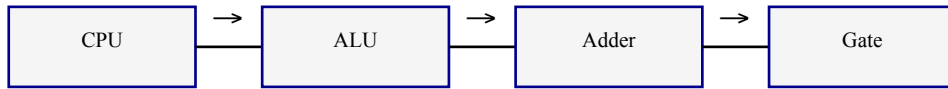
2. Inside a Modern Chip

CPU, Memory, Cache, Interconnects, I/O controllers, Billions of transistors.



3. CPU Breakdown

CPU contains Control Unit, Registers, ALU and Cache.



4. Inside the ALU

Arithmetic and Logic operations are implemented using Half Adders, Full Adders, Multiplexers, Decoders, Encoders, Comparators.



5. Full Adder

A Full Adder can be built from two Half Adders and one OR gate.



6. Half Adder

A Half Adder consists of XOR and AND gates.



7. Logic Gates

AND, OR, NOT, NAND, NOR, XOR, XNOR are the foundation of digital logic.



8. Diode Logic

Diodes can implement basic AND and OR logic by controlling current flow.



9. Why Transistors?

Diodes cannot amplify or restore logic levels. Transistors solved these limitations.



10. Modern CMOS

Modern processors use billions of MOSFET transistors to build logic gates.



11. Evolution Timeline

Mechanical → Vacuum Tube → Diode → Transistor → IC → Microprocessor → AI Chips



12. Summary

Every modern AI processor ultimately traces back to simple electronic switching principles.

